Small-Model Alignment

Training, Preference Learning, and Evaluation

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Overview. I fine-tuned Qwen2.5-0.5B with supervised learning, then independently aligned it on HH-RLHF and UltraFeedback preferences. I also trained a SmolLM2 persona adapter. Both general DPO runs improved held-out preference metrics, but that improvement did not consistently carry into generation: HH DPO reached a 50.0% tie-adjusted rate against SFT on Alpaca and 46.2% on the HH bank; UltraFeedback reached 40.8% and 38.3%. The persona adapter separated all 22 validation pairs while leaving 38 of 40 greedy answers unchanged. The central finding is a gap between *learning preference likelihoods* and *producing better answers*.

1 Models, datasets, and experimental scope

The completed project is github.com/salehfarooq/cs5326-pa2-small-model-alignment. The general pipeline uses Qwen/Qwen2.5-0.5B: base → SFT → two independent DPO policies. Both DPO policies and their frozen references start from the same SFT checkpoint, so the HH and UltraFeedback (UF) tracks differ in preference data rather than initialization. The persona experiment uses HuggingFaceTB/SmolLM2-360M-Instruct, with its unmodified instruction model as baseline and frozen reference.

Table 1: Executed training subsets. Sampling and splits use seed 42.

Track	Source and extraction	Train	Validation
SFT	HuggingFaceH4/ultrachat_200k, `train_sft`; first conversational turn	30,000	1,000
HH DPO	Anthropic/hh-rlhf; four training PM subsets, single-turn pairs	15,000	1,000
UF DPO	HuggingFaceH4/ultrafeedback_binarized, `train_prefs`	15,000	1,000
Persona	Supplied Ramsay-style preference file; all 220 rows retained	198	22

Preprocessing checks schemas and complete text, removes duplicate preference prompts and identical responses, and excludes HH multi-turn examples. Before sampling, it removes exact canonical prompt matches to the full 805-instruction Alpaca source and the supplied 300-row HH bank. General training retains complete prompts up to 256 tokens and truncates responses within a 512-token context, appending EOS before truncation.

Evaluation uses a fixed seed-42 subset of 300 Alpaca prompts and all 300 HH rows. The HH bank has 185 canonical instruction clusters; repeated rows keep their original IDs. Persona uses a separate, prompt-disjoint 40-prompt bank with ten prompts each for cooking, coding, general help, and safety.

Checkpoint scope. The reported comparisons use SFT update 938, HH update 400, UF update 469, and persona best update 5. Persona final update 13 is an additional diagnostic. Later higher-LR persona and $\beta = 0.2$ HH follow-ups completed, but their final quality evaluation is incomplete; their checkpoints are therefore outside these comparisons.

2 Training objective and executed configuration

2.1 Response-only learning

For instruction x and response y, the causal sequence score is

$$\ell_\theta(y \mid x) = \sum_{t=1}^{T} m_t \log p_\theta(z_t \mid z_{<t}), \tag{1}$$

where z is the tokenized prompt–response sequence and $m_t = 1$ only for retained response tokens, including EOS when retained. The implementation shifts predictions and targets by one position; prompt and padding positions are masked. SFT minimizes response-token mean negative log-likelihood. DPO uses the *sum* in Eq. (1), rather than a length-normalized response score.

For preferred and rejected responses y_w, y_l, define the reference-relative reward $r_\theta(x, y) = \beta[\ell_\theta(y \mid x) - \ell_{\text{ref}}(y \mid x)]$. The custom DPO loss follows [1]:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_w,y_l)} \log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l)) . \tag{2}$$

The reference stays frozen and is scored without gradients. Validation reports mean DPO loss, strict reward accuracy $\Pr(r_w > r_l)$, and mean margin $\mathbb{E}[r_w - r_l]$. Identical initial policy and reference yield loss $\log 2$, zero margin, and **0% strict accuracy**, since tied rewards do not satisfy $r_w > r_l$.

Table 2: Actual configurations of the evaluated checkpoints. All tracks use AdamW, clipping at 1.0, one epoch, and seed 42. WD denotes weight decay; $b \times a$ is microbatch size times accumulation.

Track	Trainable weights	LR / schedule / WD	$b \times a$; precision	Context; updates	β; reference; selected
SFT	Full model	2×10^{-5}; cosine, 3% warmup; 0.1	$2 \times 16 = 32$; FP32 + FP16 AMP	512; 938	–; Qwen base origin; 938
HH	Full model	10^{-6}; fixed, no warmup; 0	$1 \times 32 = 32$; FP32 + FP16 AMP	512; 469	0.1; frozen SFT 938; 400
UF	Full model	10^{-6}; fixed, no warmup; 0	$1 \times 32 = 32$; FP32 + FP16 AMP	512; 469	0.1; frozen SFT 938; 469
Persona	LoRA: $r = 8$, $\alpha = 16$, dropout 0; `q_proj`, `v_proj`	10^{-5}; fixed, no warmup; 0	$2 \times 8 = 16$; FP32, no quantization	1024; 13	0.1; SmolLM2, adapters disabled; 5

SFT and general DPO use custom PyTorch loops. Persona uses TRL's `DPOTrainer` and PEFT LoRA [2, 3]. Qwen training uses gradient checkpointing and disables the KV cache. During DPO, the policy runs on one T4 and the independent frozen reference on the second. The final 24-pair DPO accumulation window is normalized by its actual pair count; SFT accumulation is weighted by response-token count.

2.2 Prompt and generation contracts

Qwen uses the following literal template in training and evaluation, without applying its stored chat template:

```
Instruction:\n{instruction}\n\nResponse:\n
```

Persona uses SmolLM2's native chat template, including its default helpful-assistant system text. No Ramsay instruction is added by the caller. Matched evaluation is greedy: `do_sample=False`, 256 new tokens, repetition penalty 1.1, and no n-gram block. Temperature and top-p are inactive; no explicit length penalty is supplied. Qwen uses EOS/padding ID 151643. Each comparison preserves the complete instruction and uses identical decoding across models.

3 Parameter choices and deviations

Memory-driven choices. SFT microbatch 2 with accumulation 16 and DPO microbatch 1 with accumulation 32 both retain effective batch 32. The memory layout retains the guide's effective optimization batch.

Decoding-driven choices. An initial 4-gram block broke legitimate dictionary-counting code. Removing it restored executable counting functions in both base and SFT. Raising the repetition penalty from 1.05 to 1.1 subsequently stopped an observed UF development loop, but UF's planet-name length function still regressed. All final compared outputs were regenerated under 1.1/block 0. This was a documented tradeoff: improved local loop control with incomplete code preservation.

Optimization choices. General DPO keeps the guide's $\beta = 0.1$, fixed LR 10^{-6}, and one epoch. Weight decay 0 and validation every 50 updates are implementation choices. An earlier 200-update HH pilot improved some answers but failed a safety review. The full one-epoch runs were separate guide-recipe executions; the pilot did not establish an optimal setting.

Persona capacity and coverage. Rank 8, alpha 16, and query/value projections give 819,200 trainable parameters, a small adapter for 220 pairs. Zero dropout makes reference comparisons deterministic. LR 10^{-5} and one epoch limit optimization on that small dataset. Explicitly requesting 13 updates corrects partial-batch rounding

in the pinned Trainer and completes one epoch. Although context is 1024, the longest formatted preference sequence is only 123 tokens, so training pairs are not truncated. The weak generated style shows that these conservative choices did not achieve the intended persona effect.

4 Training behavior and checkpoint selection

4.1 SFT and general DPO

SFT completed 938 optimizer updates with no skips. Its validation cross-entropy initially rose from 1.374749 to 1.403624 at update 200, then declined to 1.348410 at update 938, a reduction of 0.026339 (1.92%) from initialization. The terminal checkpoint also has the lowest validation loss. The curve shows early deterioration followed by recovery, rather than uniform improvement.

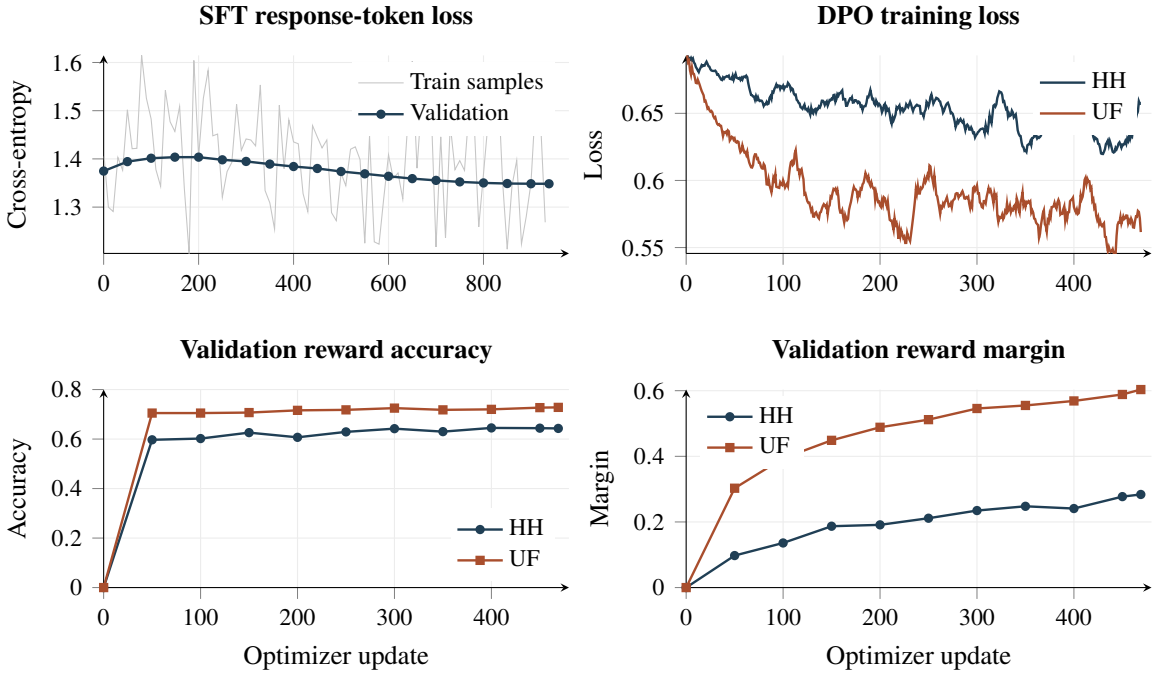


Figure 1: Recorded training behavior. SFT train-loss samples and all validation points are unsmoothed. DPO train loss uses a trailing 25-update mean; validation diagnostics are unsmoothed. Reward accuracy is exactly zero for the initially identical policy and reference.

Both DPO runs completed 469 successful updates over 15,000 pairs, with no skipped updates, finite gradients, and unchanged reference fingerprints. Table 3 shows why selecting by preference accuracy matters. HH final loss is lower than selected loss, yet its final accuracy is slightly lower; UF selects its final checkpoint.

Table 3: General DPO checkpoint selection. Loss, accuracy, and margin are from validation.

Track	Selected / final	Selected loss	Selected accuracy	Margin	Final accuracy
HH	400 / 469	0.630991	64.5%	0.240880	64.3%
UF	469 / 469	0.553575	72.8%	0.603458	72.8%

HH final loss is 0.622583. Positive reward margins establish that the policies move toward chosen responses relative to their references. They are likelihood diagnostics; their relationship to generated quality is tested separately below.

4.2 Persona preference learning

Persona validation reaches 100% strict accuracy at update 5 and ties that accuracy at updates 10 and 13. The selection rule retains the first maximum, update 5. Validation loss falls from 0.693147 to 0.690129 at the selected checkpoint and 0.685189 at the final checkpoint; the corresponding margins are 0.006046 and 0.015986. Immutable base-weight fingerprints match before and after training. The adapter separates preferences with

modest likelihood movement, despite frozen base weights.

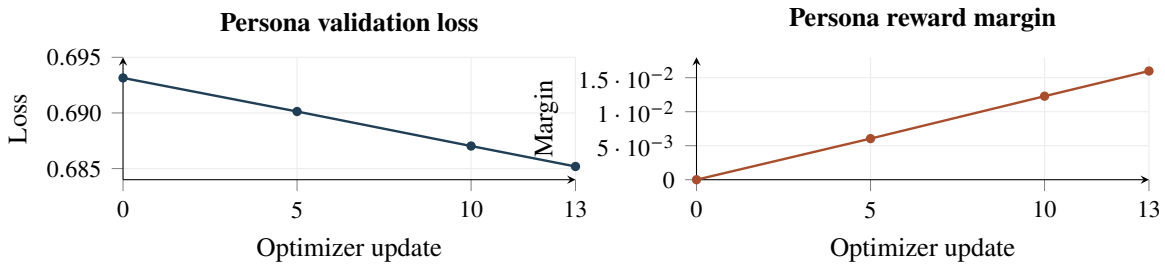


Figure 2: Original persona adapter: unsmoothed validation loss and reward margin. Accuracy first reaches its maximum at update 5, which is selected.

5 Matched evaluation results

5.1 Blind DPO–SFT comparisons

Generation persisted 2,100 responses: base on Alpaca, plus SFT and both DPO models on Alpaca and HH. All banks contain 300 answers with zero blanks. Four 300-pair files compare DPO against SFT with seed-42 A/B randomization and separate identity mappings. Local checks reconstruct the pairs byte for byte.

A separate Kaggle utility notebook judges the saved pairs with google/gemini-3.6-flash (SDK 0.6.1), using correctness, helpfulness, instruction following, formatting, and absence of hallucinations or unsafe content. The evaluator submits five pairs per request, waits at least five seconds between attempts, allows three attempts per batch, persists validated judgments after each batch, and supports resume. It reports zero unknown, unresolved-failure, and pending judgments. The recovered attempt journals record 250 request attempts for 240 initial batches: one HH/Alpaca retry and nine UF/Alpaca retries. Malformed or partial returns are validated by ID; valid judgments are retained, remaining IDs are retried, and unresolved failures are logged separately. All 1,200 judgments and the reported bootstrap intervals reproduce locally from the original saved records.

For DPO wins W, ties T, and losses L, the two rates are

$$\widehat{p}_{\text{tie}} = \frac{W + T/2}{W + T + L}, \qquad \widehat{p}_{\text{decisive}} = \frac{W}{W + L}. \tag{3}$$

Both describe DPO relative to SFT. The tie-adjusted rate has parity at 50% and is the primary comparison; unlike evaluation banks are reported separately.

Table 4: Saved blind evaluation. Each row contains $N = 300$ valid judgments. Intervals apply to the tie-adjusted rate.

DPO / bank	W	T	L	Tie-adjusted	Decisive	95% interval
HH / Alpaca	101	98	101	50.0%	50.0%	45.3–54.5%
HH / HH	115	47	138	46.2%	45.5%	38.9–53.2%
UF / Alpaca	91	63	146	40.8%	38.4%	36.2–45.7%
UF / HH	103	24	173	38.3%	37.3%	31.0–45.7%

HH DPO does not establish an improvement. It has exactly as many wins as losses on Alpaca. On HH, its point estimate is 3.83 percentage points below parity, but both HH intervals include 50%. The evidence is consistent with parity; it does not establish equivalence.

SFT outperforms UF under this judge. UF is 9.17 points below parity on Alpaca and 11.67 below on HH, with both intervals wholly below 50%. This is the clearest result across the general tracks: UF has the higher validation preference accuracy (72.8%), yet its sampled answers are judged worse.

The saved uncertainty calculation uses 2,000 bootstrap resamples and seed 42. HH resamples its 185 canonical instruction clusters instead of treating all 300 rows as independent. These intervals represent sampling variation in these comparisons, rather than judge bias or training-seed variability.

5.2 Persona style, usefulness, and safety

The original selected adapter changes only two of 40 greedy answers; the final adapter changes six. Table 5 shows that the perfect preference score does not produce a corresponding behavioral shift. No coding answer changes. All three models reach the 256-token limit on 22 of 40 prompts, and their wording remains generic rather than distinctly Ramsay-like.

Table 5: Persona results on the same 40 prompts and decoder. Output identity is a behavioral diagnostic, not a quality score.

Adapter	Val. loss	Val. accuracy	Reward margin	Identical to baseline
Best, update 5	0.690129	100%	0.006046	38/40
Final, update 13	0.685189	100%	0.015986	34/40

Qualitative checks do not show convincing persona transfer. A changed salmon answer adds slightly better closing advice while retaining unsafe doneness guidance. A changed cake-rescue answer remains unsound, and a reworded correlation/causation answer remains confused. Coding errors and harmful responses to allergy-prank and password-theft requests persist. This assessment is matched but unblinded; there is no external persona win rate. The adapter's effect is limited, and safety preservation is not established.

6 Interpretation and representative errors

6.1 What changes in the answers?

The examples below come from the frozen development bank with the *final* penalty 1.1/block-0 decoder. They illustrate aggregate findings without substituting for the 300-pair blind comparisons.

Table 6: Matched development examples: improvement, regression, and unchanged behavior. Full outputs are retained in `evaluation/*-development.json`.

Task / expected behavior	SFT	HH DPO	UF DPO
Beads: $18 + 12 - 6 + 3 = 27$	Gives 27, but calls a total a blue-bead count	Gives 9; loses the counts and bead types	Gives 27 with consistent color accounting
Bat and ball: 110 cents total, bat 100 cents more; ball costs 5	Gives 5 with correct algebra	Writes $2x + 100 = 110$, then gives 50	Gives 5 with correct algebra
Extract Maya's Friday booking and Omar's Monday invitations	Both actions	Both actions	Both actions
Integer-count function: expected `{2:3, -3:2}`	Correct function; wrong example comment	Correct function; wrong example comment	Correct function; wrong example comment
Planet-name lengths: independent executable cases	3/3 pass	3/3 pass	Counts occurrences; 1/3 pass

UF improves the bead explanation without improving the already correct numeric answer. HH regresses on both arithmetic examples; the bat-and-ball answer shows how a correct-looking equation can coexist with a wrong solution. Meeting extraction is unchanged across the final-decoder models, even though an older SFT decoder omitted an action. The integer-count example separates executable correctness from the model's claimed output.

Persona coding outputs are unchanged: first-unique-character fails `aab`, `abc`, and `aabb`, while binary search passes four edge cases. A 90-minute plan reaches 135 minutes, and a library summary invents closing times.

6.2 Why might preference learning fail to improve generation?

Observed preference-data risks. Complete-pair reviews found questionable chosen labels in both sources, including factual errors and mismatched refusals. Raw-source checks found no converter label swap. UF chosen responses are truncated in 3,987/15,000 training pairs, versus 28/15,000 for HH; the sampled mean chosen/rejected length ratios are 1.96 and 1.60. The response-summed DPO objective can therefore learn from noisy or unevenly retained comparisons. This is a plausible contributor, not an isolated causal explanation.

Observed generation risks. UF hits the response cap on 138/300 Alpaca and 163/300 HH answers, compared with SFT's 56 and 52 and HH's 41 and 34. Length limits alone do not imply poor quality. Manual inspection also confirms sustained repetition in final banks: repeated "please" in a landlord-message rewrite, hashtag repetition, and an HH emoji loop. The development decoder repair did not eliminate benchmark degeneration.

Together, these observations suggest that higher reference-relative likelihood separation is insufficient when preference labels, response coverage, and autoregressive behavior remain problematic. UF's worse blind scores are measured; data mismatch, over-optimization, and decoding sensitivity are hypotheses about their cause. The persona result is consistent with under-adaptation: small positive margins change pair ranking while rarely moving greedy decoding to a different answer. These runs do not identify which single adjustment would resolve either failure.

7 Limitations, reproducibility, and conclusion

All training uses one seed, and general evaluation uses one model judge. Exact-text exclusions do not rule out semantic overlap. The HH bank repeats instructions; the persona assessment has only 22 held-out preference pairs and 40 purposefully adversarial prompts. Decoder selection has a documented code-quality tradeoff. These limits constrain the size and generality of the quality claims.

Reproducibility. The project retains code, prepared splits, training metrics, checkpoint states and hashes, all matched outputs, blinded pairs and mappings, and the complete saved judge journals (Table 7). Its assembled code passed 381 offline tests, including all seven unchanged public adapter tests. Large weights remain in the private Kaggle outputs and were not rehashed locally during report preparation. Training does not resume optimizer state; generation and judging persist progress.

Table 7: Compact evidence index. Paths are relative to the project root.

Evidence	Local record / saved experiment
Data and configuration	`data/general/`, `configs/`, `evidence/data_summary.json`
SFT and original DPO	`evidence/sft-launch_status.json`, `evidence/dpo/`; versions 356840240 and 357186707
Matched generation and blinding	`evaluation/matched/`; version 357326672
Judge counts and intervals	`evaluation/summary.json`, `evaluation/judged/`; version 357378242
Persona baseline, best, and final	`evidence/persona-best/`, `evidence/persona-final/`; versions 357420502 and 357446665
Source and verification	`submission_adapter.py`, `src/`, `tests/`, `verify_submission.py`, `AI_USAGE.md`

Executed stacks are pinned in the project: Qwen used PyTorch 2.11.0+cu128 and Transformers 4.57.6 on two T4s; persona used Transformers 4.51.3, TRL 0.19.1, and PEFT 0.15.2. Run records retain exact model revisions and weight hashes. The completed public project is `https://github.com/salehfarooq/cs5326-pa2-small-model-alignment`. The README attributes the course starter. Assistance is disclosed in `AI_USAGE.md`.

Conclusion. The strongest result is that learning the preference objective did not reliably improve generated quality. HH DPO shows no clear gain over SFT; UF DPO performs worse under the saved judge on both banks. Persona reaches perfect validation preference separation with little useful style transfer. SFT remains the stronger baseline against UF, while preference accuracy should be interpreted alongside matched generation, correctness, and safety rather than used as their substitute.

References

[1] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. NeurIPS, 2023. arXiv:2305.18290.

[2] Hugging Face. *TRL DPOTrainer documentation*, version 0.19.1. `https://huggingface.co/docs/trl/v0.19.1/en/dpo_trainer`.

[3] Hugging Face. *PEFT LoRA configuration reference*. `https://huggingface.co/docs/peft/package_reference/lora`. Executed PEFT version: 0.15.2.